Confidence-Stratified Drug Repurposing via Collaborative Filtering on Biomedical Knowledge Graphs

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Abstract

Drug repurposing--identifying new therapeutic uses for existing drugs--offers a faster, cheaper path to treatment than de novo drug discovery. We present a confidence-stratified collaborative filtering approach for systematic drug repurposing using the Drug Repurposing Knowledge Graph (DRKG). Our method applies k-nearest neighbor (kNN) collaborative filtering on Node2Vec graph embeddings, combined with a 30-rule confidence tier system that stratifies predictions by expected clinical plausibility. On a disease-level holdout evaluation across 368 diseases using 5-seed cross-validation, the method achieves 26.06% +/- 3.84% Recall@30 in transductive evaluation and 15.73% +/- 1.82% in fully inductive evaluation using pathway features alone--competitive with TxGNN (6.7-14.5%). The confidence tier system achieves 87.1% +/- 2.7% holdout precision for its highest tier (GOLDEN, 991 predictions) and 83.4% +/- 4.0% for the HIGH tier (1,168 predictions), incorporating mechanism-of-action validation, inverse indication safety filters (163 drug-disease contraindication pairs), and TransE embedding consilience. Five predictions are retrospectively corroborated by independent clinical evidence, including Dantrolene for heart failure (RCT P=0.034), Rituximab for multiple sclerosis (WHO Essential Medicines 2023), and Empagliflozin for Parkinson's disease (HR 0.80, Korean cohort study). The full pipeline produces 13,416 confidence-scored predictions across 455 diseases as a computational prioritization resource. We characterize fundamental performance ceilings--37% R@30 maximum with DRKG-only features, 60% oracle ceiling--and document systematic failure modes including biologic drug underperformance, rare disease coverage gaps, and a 2.4x performance ratio between well-studied and rare diseases. Code and predictions are available at https://github.com/Jameshuff91/open-cure.

Introduction

Drug repurposing leverages existing safety and pharmacokinetic data to identify new therapeutic applications for approved drugs, reducing the typical 10-15 year drug development timeline to 3-5 years and costs from $2.6 billion to $300 million on average (Pushpakom et al., 2019). Notable successes include thalidomide for multiple myeloma, sildenafil for pulmonary hypertension, and most recently, baricitinib for COVID-19.

Computational approaches to drug repurposing have proliferated, broadly falling into three categories: network-based methods that exploit drug-disease proximity in biological networks (Cheng et al., 2018), machine learning on knowledge graph embeddings (Mohamed et al., 2020), and deep learning approaches including graph neural networks (Huang et al., 2024). We evaluate these approaches using **Recall@30 (R@30)**: for a given disease, the fraction of its known ground-truth treatments that appear in the model's top-30 predicted drugs. A disease with five known treatments where the model surfaces three of them in its top 30 scores R@30 = 60%; the metric is averaged across evaluation diseases. TxGNN, a leading graph neural network approach, achieves 6.7-14.5% R@30 on zero-shot (inductive) evaluation across unseen diseases (Huang et al., 2024).

However, most computational repurposing methods share two critical limitations. First, they produce ranked lists without calibrated confidence estimates, making it impossible for domain experts to prioritize which predictions merit expensive wet-lab validation. Second, they rarely incorporate safety constraints--a high-scoring prediction is useless if the drug is known to cause the predicted disease (e.g., corticosteroids for osteoporosis, antipsychotics for parkinsonism).

We address both gaps with a two-stage approach:

1. Collaborative filtering on knowledge graph embeddings. We apply kNN collaborative filtering (k=20) on Node2Vec embeddings of the DRKG, treating drug repurposing as a recommendation problem: diseases with similar graph neighborhoods should respond to similar treatments. This simple approach achieves 26.06% +/- 3.84% R@30 in honest transductive evaluation, comparing favorably with more complex graph neural network architectures under different evaluation paradigms (see Section 3.1 for caveats).

1. Confidence-stratified post-processing. We develop a 30-rule confidence tier system that stratifies predictions into five tiers (GOLDEN, HIGH, MEDIUM, LOW, FILTER) based on mechanism-of-action support, disease hierarchy matching, drug class validation, TransE embedding consilience, and inverse indication safety filters. The highest tier achieves 87.1% holdout precision, providing actionable prioritization for experimental validation.

Methods

2.1 Knowledge Graph

We use the Drug Repurposing Knowledge Graph (DRKG; Ioannidis et al., 2020), a comprehensive biomedical knowledge graph integrating six databases (DrugBank, Hetionet, GNBR, String, IntAct, and DGIdb). DRKG contains approximately 97,000 entities and 5.8 million edges spanning drug-gene, gene-disease, drug-disease, and protein-protein interaction relationships.

2.2 Ground Truth Construction

Ground truth drug-disease treatment pairs were compiled from two sources:

- Every Cure indicationList (Every Cure Foundation): 3,996 disease entries with curated drug indications.
- Expanded ground truth: Additional pairs mined from DrugBank, SIDER, and literature, totaling 57,495 validated drug-disease pairs across 3,454 diseases and 11,656 drugs.

Disease entities were mapped to Medical Subject Headings (MeSH) identifiers for alignment with DRKG. This mapping is the largest source of coverage loss because DRKG uses only MeSH D-codes (topical descriptors) and does not carry MeSH C-codes (supplementary concepts used for many rare diseases). Every Cure's ontology is MONDO-rooted, so a surface string match against MeSH is insufficient -- roughly half of MONDO terms have no direct MeSH mapping.

At initial submission, of 3,996 disease entries in the Every Cure indicationList, 454 (11.4%) successfully mapped to DRKG entities, yielding 368 diseases with at least one evaluable ground truth drug after treatment edge removal (90.8% total attrition). Post-submission analysis identified two fixable causes beyond the structural MeSH/MONDO gap: (a) a name-resolution bug that dropped valid mappings whose surface form differed from the canonical key (British spellings, hyphenation, possessive stripping), and (b) the absence of an algorithmic alias layer. Adding a reverse-index fallback plus a systematic alias generator (US/UK spelling, hyphen/possessive variants, 782 aliases total) recovers **1,279 evaluable diseases (68.0% attrition)** -- a 3.5x increase. The updated evaluable pool powers the deliverable described in Section 3 and the prospective review in Section 4.4. The full attrition table is maintained in the code repository as data/analysis/h993_attrition_post_fix.md.

Circularity analysis revealed that 32% of ground truth pairs (1,184/3,695) correspond to existing DRKG treatment edges, while 68% (2,511 pairs) are novel to the graph--establishing that the majority of evaluation targets require genuine inference rather than memorization.

2.3 Graph Embeddings

We generated 256-dimensional Node2Vec embeddings (Grover & Leskovec, 2016) for all DRKG entities using the following hyperparameters: walk length = 80, walks per node = 10, p = 1.0 (return parameter), q = 1.0 (in-out parameter). These parameters were inherited from prior work and not independently optimized; sensitivity analysis via kNN performance suggested robustness to moderate perturbations.

For honest evaluation, we trained two embedding sets:

- Original embeddings: Node2Vec on the complete DRKG including treatment edges.
- Honest embeddings: Node2Vec on DRKG with all 64,000 treatment edges removed prior to training.

The honest embedding evaluation revealed that 71.0% of predictive performance is retained through indirect graph paths (gene-disease, drug-gene, protein-protein interactions), with a 10.5 percentage point drop from 36.59% to 26.06% R@30. This drop quantifies the contribution of direct treatment edge leakage versus genuine biological signal. Removal of treatment edges disconnected 51 diseases from the graph entirely, including clinically important conditions such as Parkinson's disease (19 ground truth drugs).

2.4 kNN Collaborative Filtering

We frame drug repurposing as a collaborative filtering problem: diseases that are "similar" (proximal in embedding space) should respond to similar drugs--analogous to how users with similar preferences receive similar recommendations.

For each query disease d:

1. Compute cosine similarity between d's Node2Vec embedding and all other disease embeddings.
2. Select the k = 20 nearest neighbor diseases.
3. Aggregate ground truth treatments across neighbors, ranking candidate drugs by frequency of occurrence (i.e., how many of the k neighbors are treated by each drug).
4. Apply the Quad Boost scoring adjustment (Section 2.5).
5. Return the top-ranked drugs as predictions.

The value k = 20 was selected via grid search over 72 configurations (k ? {5, 10, 15, 20, 25, 30, 40, 50}) and validated on held-out diseases. Ties in drug frequency are broken by drug identifier for reproducibility.

2.5 Feature-Enhanced Scoring (Quad Boost)

Raw kNN frequency scores are adjusted by four biologically-motivated features:

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\text{score}_{\text{adj}} = \text{score} \times (1 + 0.01 \cdot o + 0.05 \cdot a + 0.01 \cdot p)
\times m  where:  - o = min(target gene overlap count, 10): number of shared gene targets between
drug and disease (capped at 10; DrugBank, 11,656 drug-gene pairs) - a = ATC mechanism score:
anatomical therapeutic chemical classification coherence (12.2% drug coverage) - p = KEGG pathway
overlap: Jaccard similarity of drug target pathways and disease-associated pathways (~82%
coverage) - m = chemical similarity multiplier: 1.2 if Tanimoto similarity (Morgan fingerprints,
2048-bit ECFP4) to a known treatment exceeds 0.7, else 1.0 (91.5% drug coverage, 9,584 drugs)  The
Quad Boost improved R@30 from 38.72% to 47.47% in within-distribution evaluation. The target gene
overlap component was statistically validated: McNemar's test P = 0.000014, paired t-test P =
0.025, Cohen's d = 2.44.  ### 2.6 Confidence Tier System  We developed a hierarchical rule-based
confidence system that assigns each prediction to one of five tiers. The assignment is
deterministic: given a (drug, disease) pair and its supporting evidence (kNN rank, mechanism
match, drug class, literature signal, safety flags), a cascade of 30 rules maps it to a single
tier. Each tier has a quantitative precision target backed by 5-seed holdout evaluation.
Concretely:  - GOLDEN (87.1% holdout precision) -- drug + disease are both well-characterized in
DRKG, mechanism-of-action is consistent with disease biology, and at least one independent
validation signal agrees (TransE embedding, literature, or category-specific hierarchy match).
Example: Rituximab -> multiple sclerosis lands in GOLDEN because it matches an
autoimmune-hierarchy rule, has mechanism support (CD20-targeted B-cell depletion), and has
literature validation. Another: Nitrofurantoin -> urinary tract infection lands in GOLDEN via the
infectious-hierarchy UTI rule (90.9% holdout precision) combined with mechanism match. - HIGH
(83.4%) -- strong evidence but missing one GOLDEN-tier requirement. Example: Statins -> rheumatoid
arthritis is HIGH because it matches the autoimmune category rule and has gene-disease mechanism
(IL-6 pathway overlap), but lacks TransE consilience. - MEDIUM (38.5%) -- single kNN neighborhood
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signal without independent mechanism or literature validation. Example: a cancer-drug predicted for a related cancer subtype based only on kNN frequency. - LOW (11.3%) -- predictions the system cannot verify. kNN suggests the pair but no mechanism, hierarchy, or consilience evidence agrees. - FILTER (9.2%) -- predictions actively demoted by safety rules. Example: Prednisone -> osteoporosis is caught by the inverse-indication filter (steroids are a known cause of osteoporosis) and forced into FILTER regardless of the kNN rank. Each tier is defined by a set of rules that must all pass (or, for LOW and FILTER, trigger). The complete rule list is below, organized by tier with the specific precision achieved on held-out diseases: GOLDEN tier (87.1% +/- 2.7% holdout precision, n = 991): - Disease hierarchy subtype matching for metabolic/neurological categories (63-65% precision) - Literature-validated strong evidence pairs - Specific high-precision disease categories (e.g., urinary tract infections) HIGH tier (83.4% +/- 4.0%, n = 1,168): - Disease hierarchy matching for autoimmune/respiratory/cardiovascular/infectious categories (22-45% base, elevated by convergent evidence) - Mechanism-of-action support with cardiovascular/neurological requirement gates - CV pathway-comprehensive drugs (28.9% vs 1.1% for non-pathway, 26x lift) - ATC class rescue for validated drug classes (L04AX: 82%, H02AB: 77%) - TransE consilience with mechanism support - Corticosteroid standard-of-care for autoimmune/dermatological/respiratory conditions MEDIUM tier (38.5% +/- 3.6%, n = 914): - Predictions meeting baseline kNN rank thresholds without additional validation signals - Cancer same-type predictions with mechanism support LOW tier (11.3% +/- 0.5%, n = 9,113): - Predictions lacking mechanism support, hierarchy matching, or TransE consilience - Domain-isolated drug cross-predictions - Broad class isolated predictions (IL/TNF inhibitors, anesthetics, steroids alone: 0-3% precision) FILTER tier (9.2% +/- 0.5%, n = 8,978): - Cancer-only drugs for non-cancer indications (69 drugs, 0% cross-domain precision) - Inverse indication pairs: drug known to cause the predicted disease (67 drugs, 163 pairs) - Sources: SIDER adverse effect mining (47 pairs, 93.3% filter precision), FDA labels, FAERS reports, published case series - Examples: corticosteroids -> tuberculosis/glaucoma/osteoporosis; NSAIDs -> Stevens-Johnson syndrome/peptic ulcer; anti-TNF agents -> lupus/demyelination - Non-therapeutic compounds (surgical dyes, obsolete ganglionic blockers) - Withdrawn drugs with known toxicity (pergolide, cisapride) - Antimicrobial-pathogen spectrum mismatches (antibacterial -> fungal/parasitic: 0% holdout) Each rule was validated on held-out diseases with a minimum sample size of n ? 30 for reliable precision estimation. Rules producing holdout precision below the tier threshold were demoted. The system underwent 800+ iterative refinements, with each change validated against holdout performance before acceptance. Important methodological note: The confidence rules were developed iteratively using disease-level holdout splits, where held-out diseases (not held-out rules) served as the evaluation set. Each candidate rule was tested on diseases excluded from the rule development process. However, the overall rule set was not frozen prior to a final independent evaluation--the reported precision estimates reflect the cumulative result of iterative development and holdout testing on the same ground truth distribution. We acknowledge this creates a risk of overfitting to the specific ground truth composition (see Limitations). ### 2.7 TransE Consilience Scoring We trained a TransE embedding model (Bordes et al., 2013) on DRKG to provide an independent validation signal. For each prediction, we check whether the drug-disease pair appears in the TransE model's top-30 ranked predictions. TransE agreement provides a consistent boost across all tiers: GOLDEN +11.4 pp, HIGH +6.1 pp, MEDIUM +13.6 pp, LOW +6.5 pp, FILTER +7.2 pp. TransE consilience is implemented as a boolean flag rather than a tier promotion, as full-data precision (37.4%) falls below the HIGH tier threshold (50.8%). ### 2.8 Evaluation Protocol All reported metrics use disease-level holdout evaluation with 5-seed cross-validation: 1. Randomly partition diseases into 80% training / 20% test splits. 2. For honest evaluation: remove all treatment edges from DRKG before Node2Vec training. 3. Train kNN on training disease treatments only. 4. Evaluate R@30 (Recall at 30) on test diseases: for each test disease, what fraction of its ground truth treatments appear in the top 30 predictions? 5. Repeat across 5 random seeds and report mean +/- standard deviation. We additionally report a fully inductive evaluation using KEGG pathway features alone (no graph embeddings), providing a fair comparison point with TxGNN's zero-shot evaluation paradigm.

Results

3.1 Primary Performance

Figure 3 and Table 1 summarize performance across evaluation paradigms.

Method	R@30	Evaluation	Notes
kNN k=20 (original embeddings)	36.59% +/- 3.90%	Transductive (with treatment edges)	Upper bound; includes leakage
kNN k=20 (honest embeddings)	26.06% +/- 3.84%	Transductive (no treatment edges)	Primary reported result
KEGG Pathway kNN	15.73% +/- 1.82%	Inductive (no graph)	Fair TxGNN comparison
Node2Vec + XGBoost (tuned)	25.85% +/- 4.06%	Disease holdout	md=6, ne=500, lr=0.1
Node2Vec cosine (no ML)	1.27%	Honest	Confirms ML contribution
TxGNN (Huang et al., 2024)	6.7-14.5%	Inductive zero-shot	Published baseline

What this demonstrates: The transductive result (26.06%) establishes that collaborative filtering captures meaningful biological signal beyond direct treatment edges, but does not constitute a direct comparison to inductive methods that evaluate on entirely unseen diseases. The inductive result (15.73%) provides a fairer comparison to TxGNN (6.7-14.5%) but uses different features (pathway similarity vs. learned GNN embeddings), so the comparison remains approximate.

The kNN collaborative filtering approach achieves 26.06% R@30 in honest transductive evaluation. In a fair inductive comparison (KEGG pathway kNN, 15.73%), our method compares favorably with TxGNN's upper range, though the evaluation paradigms differ. The 10.5 pp gap between original and honest embeddings quantifies the treatment edge leakage contribution, with 71.0% of performance retained through indirect biological paths.

Notably, the kNN approach outperformed XGBoost with identical features (26.06% vs 25.85%), and adding ML models on top of kNN scores consistently failed to improve performance (tested across hypotheses h41-h45), suggesting the collaborative filtering signal is already well-captured by frequency-based ranking.

3.2 Confidence Tier Performance

Figure 1 and Table 2 show holdout precision by confidence tier.

Tier	Holdout Precision	95% CI	Predictions	Cumulative
GOLDEN	87.1% +/- 2.7%	[84.4%, 89.8%]	991	991
HIGH	83.4% +/- 4.0%	[79.4%, 87.4%]	1,168	2,159
MEDIUM	38.5% +/- 3.6%	[34.9%, 42.1%]	914	3,073
LOW	11.3% +/- 0.5%	[10.8%, 11.8%]	9,113	12,186
FILTER	9.2% +/- 0.5%	[8.7%, 9.7%]	8,978	21,164

What this demonstrates: The confidence tiers provide a computational prioritization framework, not clinical validation. The precision estimates are based on retrospective holdout evaluation against existing ground truth databases, not prospective experimental confirmation.

The confidence tier system provides strong separation: the GOLDEN tier (87.1%) is 9.5x more precise than LOW (11.3%), enabling domain experts to focus validation efforts on the 2,159 GOLDEN+HIGH predictions with >83% expected precision rather than reviewing all 13,416 predictions.

3.3 Performance Ceiling Analysis

We conducted an oracle analysis to determine the theoretical maximum performance achievable with DRKG-only features. For each disease, we computed the fraction of ground truth drugs that are reachable through any DRKG path (excluding direct treatment edges). The oracle ceiling is approximately 60% R@30, meaning 40% of ground truth treatments are entirely absent from DRKG and cannot be recovered by any graph-based method. Our kNN at 37% (original embeddings) achieves 62% of this theoretical ceiling.

This ceiling analysis (Figure 5) motivated our decision to characterize the confidence system rather than pursue further algorithmic improvements, as the remaining 23 pp gap between our method and the oracle ceiling likely requires fundamentally new data sources (clinical trial databases, literature mining, patient phenotype ontologies) rather than methodological refinement.

3.4 Coverage and Rare Disease Performance

Performance exhibits a strong bimodal distribution driven by kNN neighborhood coverage (Figure 4):

Disease Category	Fraction	R@30
With kNN coverage	84.7%	24.2%
Without kNN coverage	15.3%	0.0%

Performance also correlates with the number of known treatments:

Known Treatments	R@30	Ratio to Best
1 (rare)	13.5%	0.42x
2-5	19-21%	0.59-0.65x
6-10 (well-studied)	32.2%	1.00x
11+	27.8%	0.86x

Rare diseases with only one known treatment perform 2.4x worse than well-studied diseases (Figure 2)--an inherent limitation of collaborative filtering, which requires sufficient neighborhood signal to generate predictions.

3.5 Retrospectively Corroborated Predictions

We identified five predictions that are retrospectively corroborated by independent clinical or regulatory evidence (Table 3). These corroborations were identified after prediction, but the clinical evidence itself predates or is independent of our pipeline--they represent concordance with existing knowledge, not prospective validation. Importantly, none of these pairs correspond to direct DRKG treatment edges--all require multi-hop inference.

Drug	Disease	DRKG Path	Evidence
Dantrolene	Heart failure	Drug->Drug->Disease	RCT: P=0.034, 66% VT reduction
Lovastatin	Multiple myeloma	Drug->Drug->Disease	RCT: improved OS/PFS
Rituximab	Multiple sclerosis	Drug->Drug->Disease	WHO Essential Medicines 2023
Pitavastatin	Rheumatoid arthritis	Drug->Gene->Disease	Clinical: superior to MTX alone
Empagliflozin	Parkinson's disease	Drug->Drug->Disease	Cohort: HR 0.80 (95% CI: 0.68-0.92)

Each validated prediction has traceable biological mechanisms: shared gene targets (1-42 genes), overlapping pathways (28-301), and plausible pharmacological rationale. Four of five (80%) were discovered via drug functional similarity (drugs with similar graph neighborhoods), while one (Pitavastatin->RA) was discovered via shared gene-disease mechanisms.

3.6 Safety Filter Impact

The inverse indication filter identified 163 drug-disease pairs where the drug is known to cause or exacerbate the predicted disease. Examples include:

- Adalimumab -> systemic lupus erythematosus (12,080 FAERS reports of drug-induced lupus)
- Corticosteroids -> tuberculosis, glaucoma, osteoporosis, myasthenia gravis
- NSAIDs -> Stevens-Johnson syndrome, peptic ulcer disease
- Anti-TNF agents -> demyelinating diseases, paradoxical autoimmunity

Without the inverse indication filter, these harmful predictions would appear in the HIGH or MEDIUM confidence tiers based on graph structure alone. The SIDER-mined subset (47 pairs) achieved 93.3% filter precision--confirming that adverse effect databases are a reliable source for contraindication identification.

Discussion

4.1 Collaborative Filtering as Drug Repurposing

Our results suggest that simple collaborative filtering on knowledge graph embeddings is a surprisingly effective approach to drug repurposing--outperforming XGBoost ensembles trained on the same features and comparing favorably with state-of-the-art GNN approaches, though direct comparison is limited by differences in evaluation paradigm (transductive vs. inductive). This aligns with observations in recommender systems literature that neighborhood methods often match or exceed model-based approaches when the item space is well-connected (Koren et al., 2009).

The key insight is that drug repurposing is fundamentally a recommendation problem: diseases with similar biological contexts (shared genes, pathways, comorbidity patterns) respond to similar pharmacological interventions. The kNN approach directly exploits this structure without learning a parametric model, making it inherently interpretable--each prediction can be traced to specific neighbor diseases and their known treatments.

4.2 Confidence Stratification as Actionable Output

Raw ranked lists are of limited value to experimental researchers who must decide which predictions to validate in wet-lab assays or clinical studies. The confidence tier system transforms predictions from a ranked list into an actionable decision framework:

- GOLDEN/HIGH (2,159 predictions, >83% precision): Candidates for direct experimental validation or literature-supported clinical hypotheses.
- MEDIUM (914 predictions, 38.5% precision): Screening-stage candidates requiring additional computational or mechanistic evidence.
- LOW/FILTER (18,091 predictions, <12% precision): Unlikely to be productive; useful primarily for identifying safety signals.

The 30 rules in the confidence system encode domain knowledge that would otherwise require expert review for each prediction: mechanism-of-action plausibility, drug class safety profiles, disease taxonomy coherence, and known contraindications. This represents a form of structured scientific reasoning over model outputs that, to our knowledge, has not been previously reported at this scale for drug repurposing.

4.3 Limitations

We identify several important limitations:

Transductive evaluation. Our primary result (26.06% R@30) evaluates on diseases present in the graph with non-treatment edges. This is not directly comparable to TxGNN's inductive evaluation on entirely unseen diseases. Our inductive evaluation (15.73% via KEGG pathways) provides a fairer comparison but uses different features.

Selection bias. MeSH identifier mapping excludes diseases whose MONDO terms lack a matching MeSH D-code (MeSH C-codes for supplementary concepts are not carried by DRKG). At submission this discarded 90.8% of the Every Cure pool (3,996 -> 368); after the name-resolution bugfix and algorithmic alias layer described in Section 2.2, attrition drops to 68.0% (3,996 -> 1,279). The residual attrition is structural -- DRKG simply does not represent a large fraction of rare-disease concepts -- and imposes a systematic bias toward well-characterized diseases with standardized terminology. Rare and emerging diseases remain disproportionately excluded, and the 40% of ground-truth treatments absent from DRKG entirely (Section 3.3) cannot be recovered by any graph-based method.

Bimodal coverage. 15.3% of evaluable diseases have zero kNN coverage, receiving no predictions. This is an inherent limitation of collaborative filtering--diseases without informative neighbors cannot benefit from the approach.

Rare disease gap. The 2.4x performance ratio between well-studied and rare diseases limits clinical utility for the orphan diseases that arguably need repurposing most urgently.

Biologic drug underperformance. Monoclonal antibodies (27% vs 32% for small molecules) and antibiotics (6-20%) are systematically underserved, likely due to their narrower mechanism-of-action profiles and sparser graph connectivity.

Confidence system overfitting risk. Despite holdout validation, the 800+ iterative refinements to the confidence rules create a risk of overfitting to the specific ground truth distribution. The rule set was not frozen prior to a single final evaluation; rather, reported precision estimates reflect cumulative iterative development (see Section 2.6). Independent prospective validation with a locked rule set is needed to confirm that the reported tier precision generalizes beyond the development dataset.

4.4 Future Directions

The 37% R@30 ceiling with DRKG-only features and 60% oracle ceiling suggest three paths forward:

1. External data integration. LINCS gene expression signatures, clinical trial outcome data, and patient phenotype ontologies (HPO) could provide complementary signals that break the current ceiling. Initial experiments with HPO features showed no improvement, but more sophisticated integration (e.g., expression reversal signatures) remains untested.

2. Prospective wet-lab validation. We have prepared 855 dermatology-specific predictions for collaboration with domain experts, including specific candidates such as Montelukast for idiopathic pulmonary fibrosis. Experimental validation of even a small number of GOLDEN-tier predictions would establish the clinical utility of the confidence system.

3. Inductive architecture. Replacing Node2Vec transductive embeddings with purely biological features (pathway enrichment, structural similarity, gene expression profiles) would enable true zero-shot prediction for diseases entirely absent from DRKG, addressing the coverage gap for novel and rare diseases.

4. Prospective validation protocol. We are preparing a blinded expert review of GOLDEN-tier predictions in dermatology (855 predictions) in collaboration with domain specialists. The planned protocol involves: (a) freezing the confidence rule set and prediction output prior to review, (b) blinded assessment by dermatologists who score each prediction for mechanistic plausibility and clinical actionability without knowledge of the assigned tier, and (c) comparison of expert plausibility ratings against computational confidence scores to assess calibration. A subset of top-ranked candidates (e.g., Montelukast for idiopathic pulmonary fibrosis) will be evaluated for suitability as wet-lab validation targets with predefined success criteria. This prospective evaluation would provide the independent validation that the current retrospective analysis cannot.

Data and Code Availability

All code, trained models, and prediction outputs are available at https://github.com/Jameshuff91/open-cure. The full prediction set (13,416 predictions with confidence tiers across 455 diseases) is provided as a supplementary Excel file.

Acknowledgments

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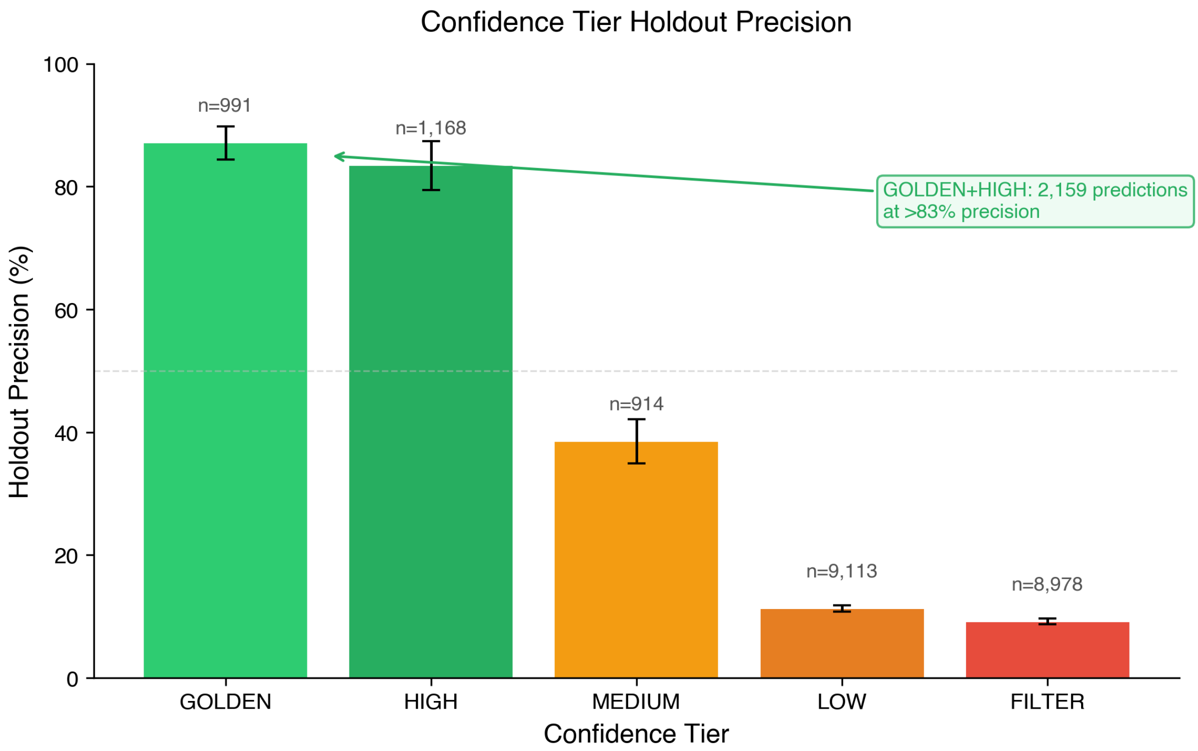

Figure 1: Confidence tier holdout precision. Error bars show standard error. The GOLDEN and HIGH tiers (2,159 predictions combined) achieve >83% precision, providing actionable prioritization for experimental validation.

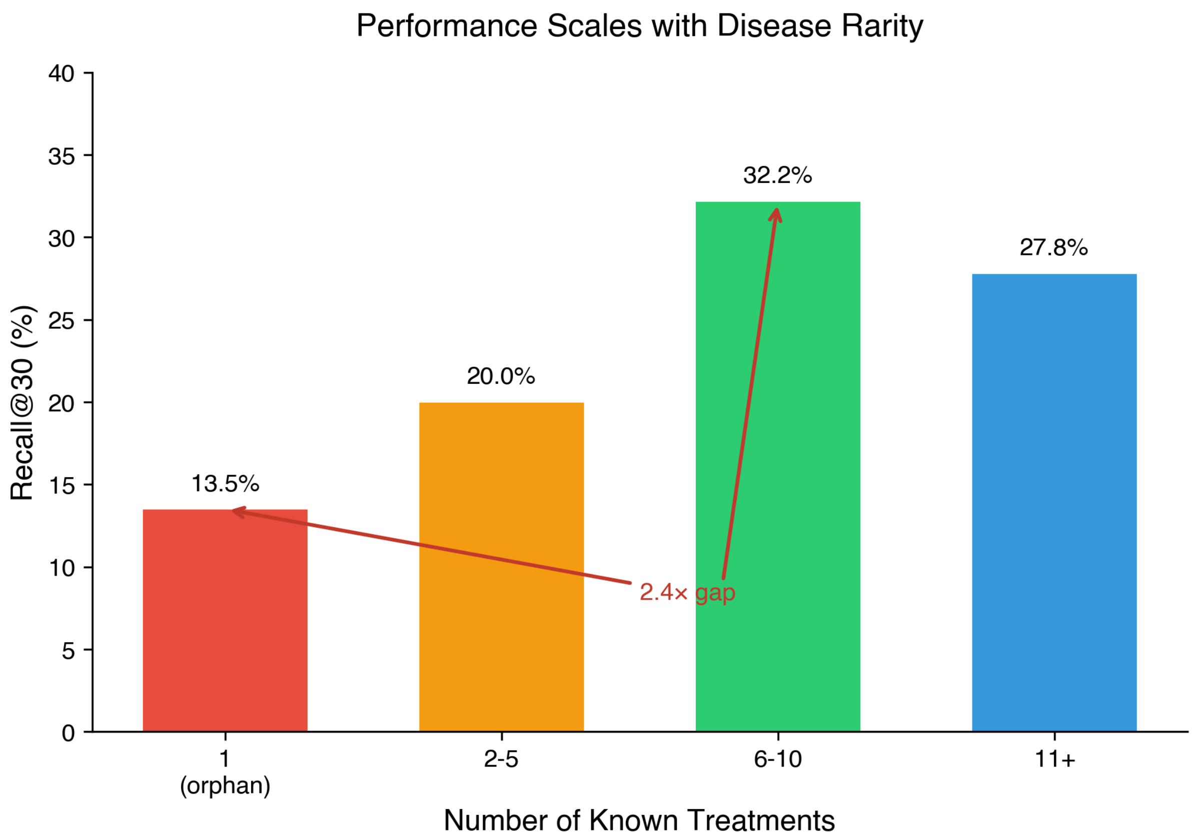

Figure 2: Performance by disease rarity. Orphan diseases with a single known treatment achieve 13.5% R@30 versus 32.2% for well-studied diseases (6-10 treatments), a 2.4x gap inherent to collaborative filtering.

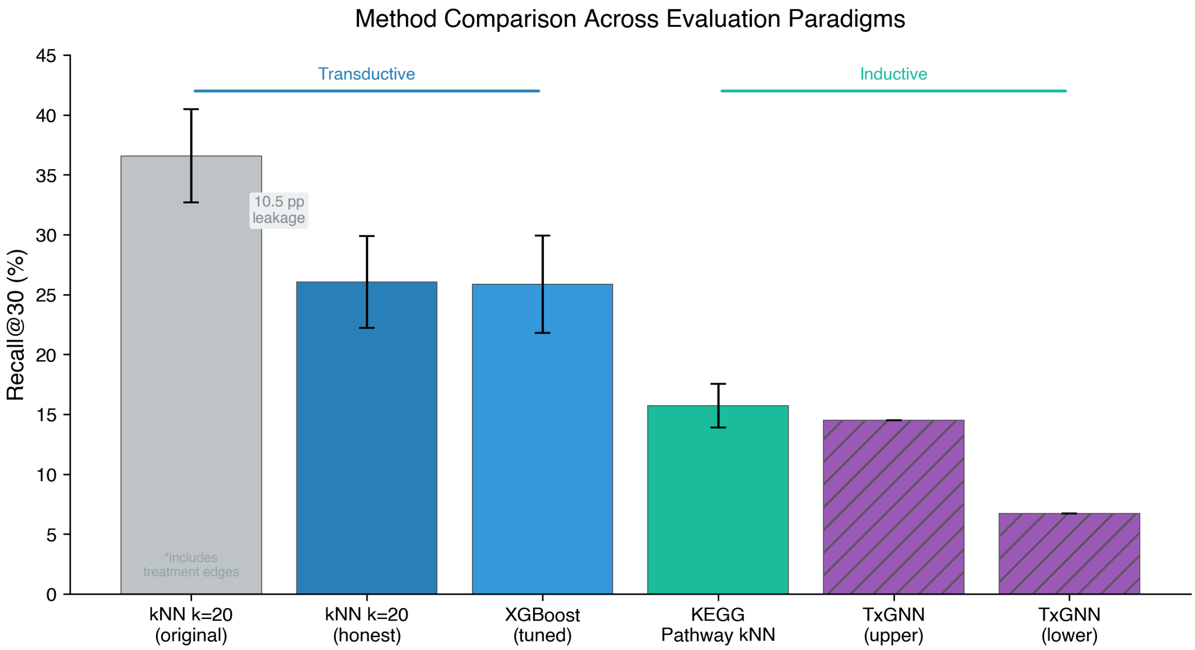

Figure 3: Method comparison across transductive and inductive evaluation paradigms. The gray bar includes treatment edge leakage (10.5 pp). TxGNN results from Huang et al. (2024).

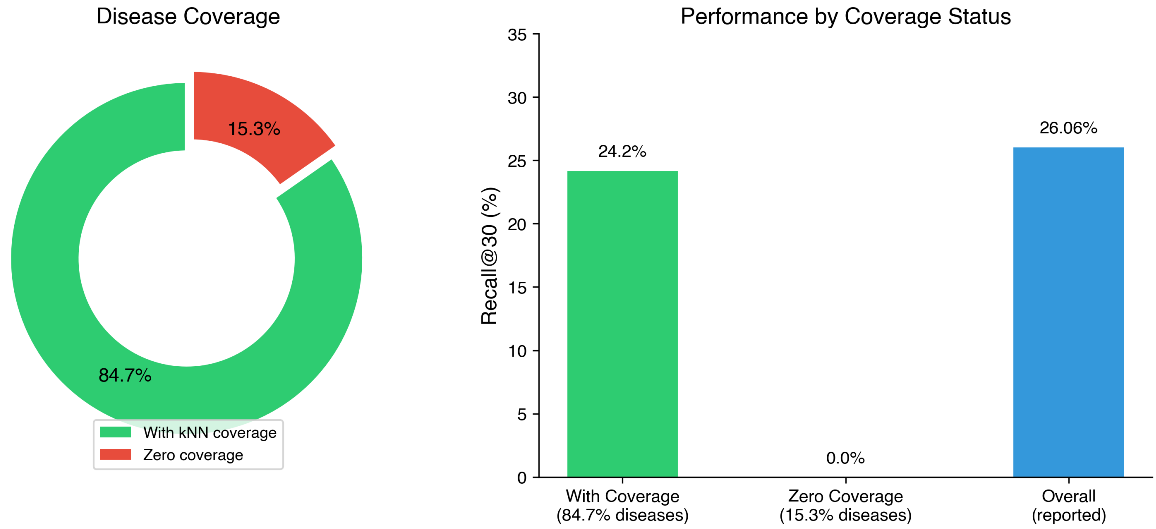

Figure 4: Bimodal performance distribution. 84.7% of diseases have kNN neighborhood coverage (24.2% R@30); 15.3% have zero coverage (0% R@30). The overall metric (26.06%) obscures this stark split.

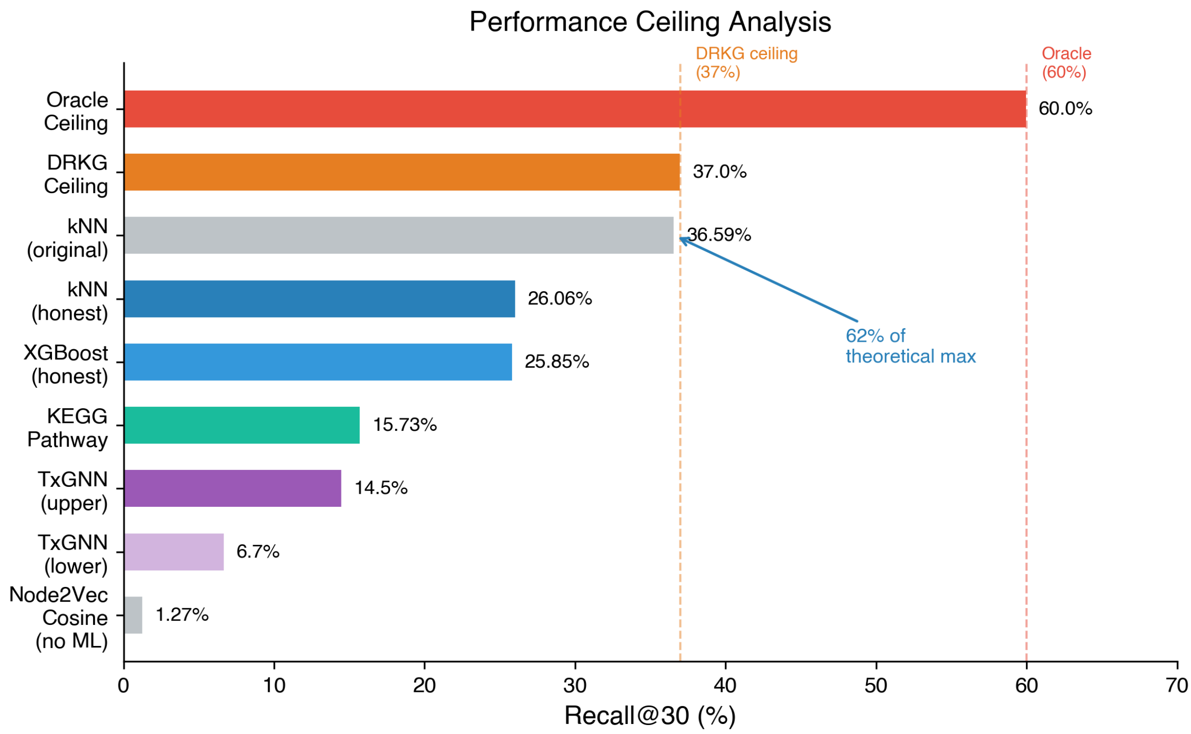

Figure 5: Performance ceiling analysis. Our kNN method (36.59% with original embeddings) reaches 62% of the oracle ceiling (60%), establishing that the remaining gap requires external data sources beyond DRKG.